

1. Introduction

The objective of this project was to design and optimize a two-stage, Miller-compensated CMOS operational amplifier using a 0.18μm process. The design goal was to balance conflicting performance metrics, including open-loop gain, unity-gain bandwidth (GBW), phase margin (PM), slew rate (SR), and power consumption. The finalized design successfully meets all specifications through iterative optimization of the bias network and compensation capacitance.

2. Description of Design

(a) Circuit Overview and Function:

The circuit (figure 1) utilizes a classic two-stage architecture:

- i) **First Stage:** An NMOS differential pair (M1, M2) with a PMOS active load (M3, M4). This stage provides the bulk of the voltage gain and establishes the Input Common-Mode Range (ICMR).
- ii) **Second Stage:** A PMOS common-source amplifier (M6) with an NMOS current source load (M7). This stage provides additional gain and allows for a wide output voltage swing.
- iii) **Bias Network:** This design uses a precision resistive divider (R_1 , R_2) to establish the gate voltage ($V_{GS5,7}$) for the NMOS current sources.

The Logic: R_1 , R_2 create a stable DC voltage $V_{GS} = VDD \times \frac{R_2}{R_1 + R_2}$. This voltage is fed directly to the gates of M₅ (tail current) and M₇ (second-stage load). By decreasing R_1 , we increased the gate voltage, which raised the V_{GS} of the transistors and effectively "turned up the volume" on the total circuit current to meet the speed requirements.

- iv) **Compensation:** A Miller capacitor (C_c) is placed across the second stage to perform "pole splitting," ensuring stability by pushing the dominant pole to a lower frequency and the non-dominant pole to a higher frequency.

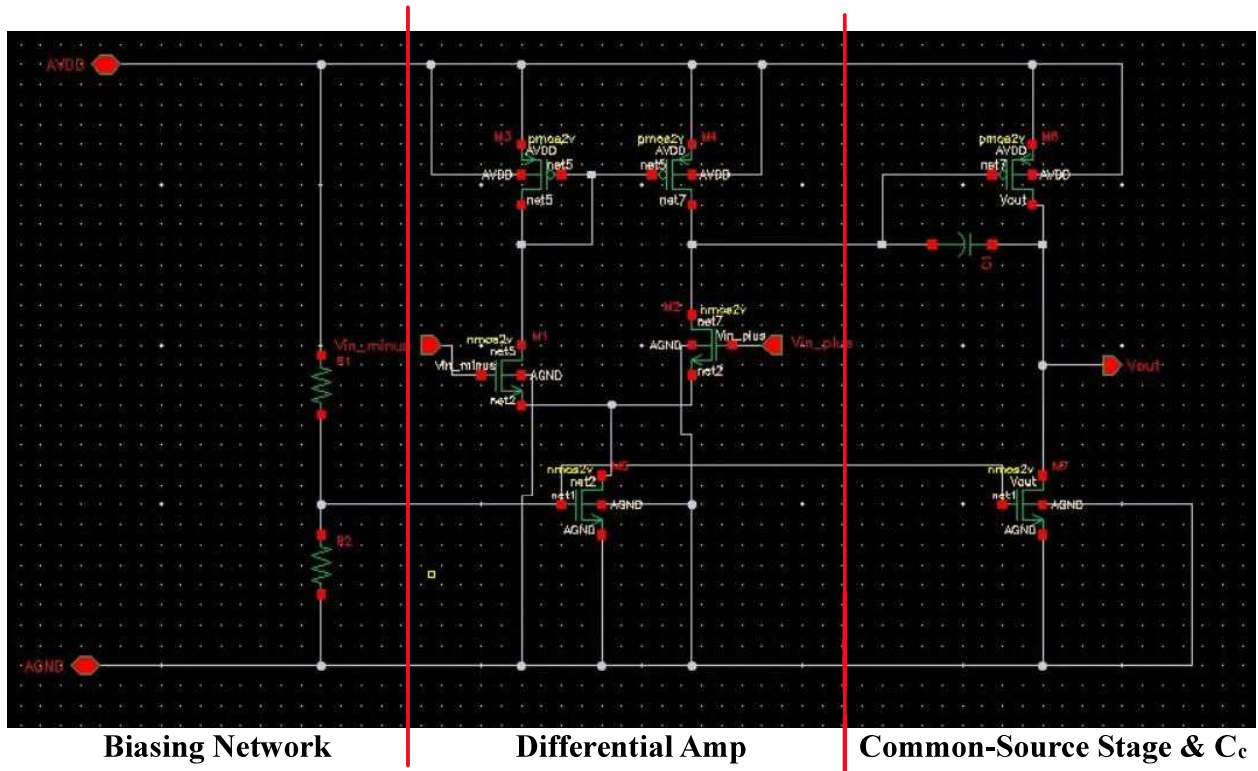


Figure 1: Design of Two-stage, Miller-compensated CMOS Operational Amplifier

(b) Hand Calculations and Methodology¹

The design process followed a top-down approach, it evolved through two distinct iterations of hand calculations to find the optimal balance between gain and frequency response:

Iteration 1: Baseline Establishment

The first iteration focused on meeting the fundamental stability and speed requirements:

- With a load capacitance of 2pF, the Miller compensation capacitor was initially set to 0.5pF.
- To meet the Slew Rate target, the tail current I_5 was calculated as $5 \mu\text{A}$
- While this setup satisfied the bandwidth requirements, the resulting open-loop gain was only 84 V/V, far below the 1000 V/V goal. This revealed that the transconductance of the stages was insufficient at the initial baseline transistor ratios. As such, the Ratios of $M_{1,2}$ and M_6 were increased. In doing so, we increased the transconductance of $gm_{1,2,6}$ thereby meeting our requirements.

¹ This is an overview highlighting the most important details, but a rigorous and detailed hand calculation is provided in Appendix A at the end of this report.

Parameter	Specification	Calculation Result	Status
C_C	$0.22C_L$	0.5 pF	Pass
I_5		$5\ \mu\text{A}$	
$S_{3,4}$ (W/L)	≤ 1000	1	Pass
M_3 Pole	$> 10 \times (10\ \text{MHz})$	340 MHz	Pass
$S_{1,2}$ (W/L)	≤ 1000	1	Pass
S_5 (W/L)	≤ 1000	2	Pass
S_6 (W/L)	≤ 1000	13	Pass
Vout Max	$\geq 1.5\ \text{V}$	1.595 V	Pass
A_v	$\geq 1000\ \text{V/V}$	84 V/V	Fail

Table 1: Results of First Calculation Iteration

Iteration 2: Gain Optimization

The second iteration focused on improving the amplifier gain by significantly increasing the transconductance of both stages.

- First Stage: The W/L ratio of S_1 and S_2 was increased from a minimum value of 1 to 130. This adjustment raised $g_{m1,2}$ to $4.1456 \times 10^{-4}\ \text{S}$, providing sufficient drive strength for the first stage.
- Second Stage: To preserve non-dominant pole separation ($g_{m6} \approx 10 \cdot g_{m2} = 4.1456 \times 10^{-3}\ \text{S}$), the size of S_6 was increased to 171 to account for the new g_{m6} .
- This iteration achieved a calculated gain of 1104 V/V (approximately 60.8 dB), successfully meeting the gain specification.

Parameter	Specification	Calculation Result	Status
C_C	$0.22C_L$	0.5 pF	Pass
I_5		$5\ \mu\text{A}$	
$S_{3,4}$ (W/L)	≤ 1000	1	Pass
M_3 Pole	$> 10 \times (10\ \text{MHz})$	340 MHz	Pass
$S_{1,2}$ (W/L)	≤ 1000	130	Pass
$g_{m1,2}$		$4.1456 \times 10^{-4}\ \text{S}$	
S_5 (W/L)	≤ 1000	2	Pass
S_6 (W/L)	≤ 1000	171	Pass
g_{m6}		$4.1456 \times 10^{-3}\ \text{S}$	
I_6		$425.23\ \mu\text{A}$	
Vout Max	$\geq 1.5\ \text{V}$	1.5948 V	Pass
S_7 (W/L)	≤ 1000	170	Pass

Vout Min	$\leq 0.4 \text{ V}$	0.137 V	Pass
A_v	$\geq 1000 \text{ V/V}$	1104 V/V	Pass
R1		212 K Ω	
R2		100 K Ω	
I_p	$\leq 5\text{mA}$	436 μA	Pass

Table 2: Detailed Results of Second Calculation Iteration

3. Simulation Results

This section presents the final performance verification of the operational amplifier. All simulations were conducted using the optimized design parameters²:

$$R_1 = 175 \text{ k}\Omega \text{ and } C_c = 0.65 \text{ pF}$$

(a) Testbench Setup

- **Open-Loop AC Analysis:**

To evaluate the frequency response without driving the amplifier into saturation, a “big-RC” feedback network was employed. This configuration uses a 1 G Ω resistor in the feedback path and a 1 F capacitor for AC coupling. It effectively breaks the loop for AC signals while maintaining a stable DC operating point.

- **DC Sweep Analysis:**

The input voltage was swept from 0 V to 1.8 V to verify the linear output swing and confirm that the tail current remains constant across the specified input common-mode range (ICMR).

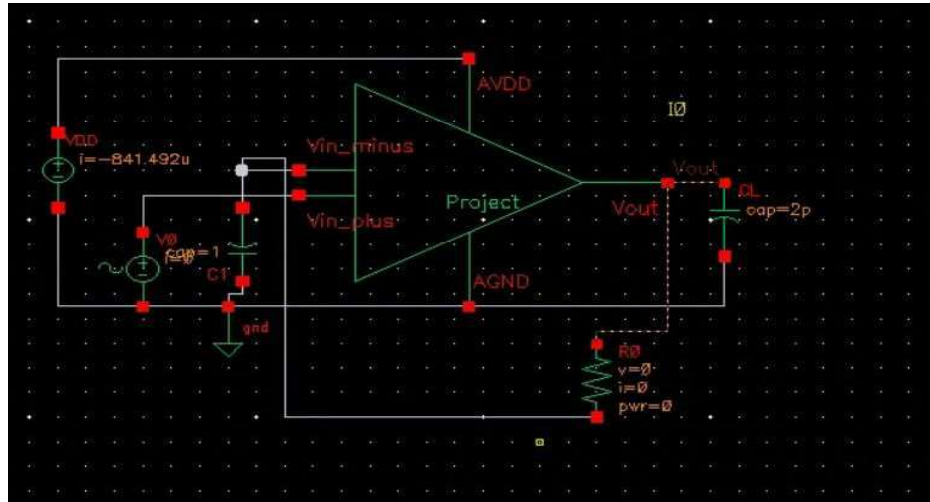


Figure 2: Testbench Setup Used for Open-Loop AC Analysis & DC Sweep³

² How these design variable were optimized and why will be further explained in the next Step. It will also be compared to hand calculations. This section will only showcase the results of the OP-Amp that meets the specification required.

³ The DC Sweep analysis was performed using the stability-compensated testbench. Due to the high-impedance resistive feedback (R0) and the negligible gate leakage of the CMOS process, the testbench maintains a perfect DC unity-gain configuration while allowing for seamless transition to AC frequency sweeps.

- **Transient Analysis (Slew Rate):**

The op-amp was configured as a unity-gain buffer. A pulse input with a 0.5 V step (from 0.7 V to 1.2 V) was applied to the non-inverting terminal to drive the circuit into a large-signal slewing condition.

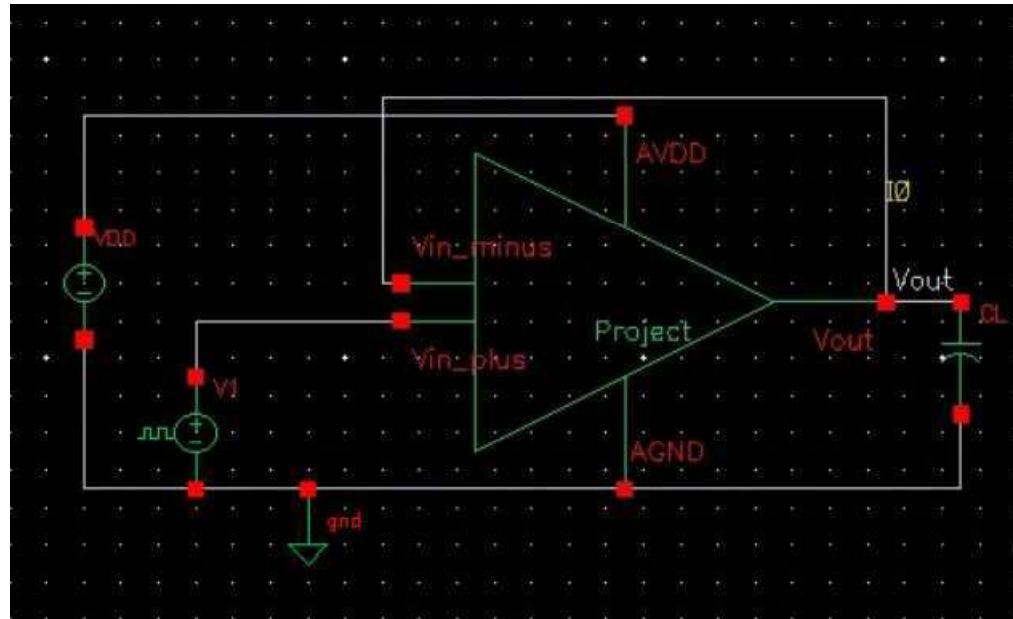


Figure 3: Testbench Setup Used for Transient Analysis

Add Instance

☒ Add Wire Stubs at:
☐ all terminals ☒ registered terminals only

Array Rows: 1 Columns: 1

Rotate Sideways Upside Down

Frequency name for 1/period:

Noise file name:

Number of noise/freq pairs: 0

DC voltage:

AC magnitude:

AC phase:

XF magnitude:

PAC magnitude:

PAC phase:

Voltage 1: 700.0m V

Voltage 2: 1.2 V

Period: 10u s

Delay time: 10n s

Rise time: 1n s

Fall time: 1n s

Pulse width: 5u s

Temperature coefficient 1:

Hide Cancel Defaults Help

Figure 4: V Pulse Setup Used for Transient Analysis

(b) Performance Plots and Verification

The following results demonstrate that the design meets all target specifications.

Open-Loop Frequency Response

The AC analysis confirms both high gain and stable operation:

- Open-Loop Gain (A_v): 78 dB (7.85448 KV)
- Unity-Gain Bandwidth (GBW): 15.94 MHz
- Phase Margin (PM):

The phase at the unity-gain frequency was measured as -119.45° , giving:

$$PM = 180^\circ - 119.45^\circ = 60.55^\circ$$

DC Response (Output Swing and ICMR)

- Output Swing: The output remains linear over the required range of 0.4 V to 1.5 V.
- Input Common-Mode Range (ICMR): The tail current (I_{M5}) remains approximately constant at 9.43 μA across the input sweep, confirming that the input differential pair operates in saturation.
- The DC operating points of all MOS transistors were examined to confirm operation in the saturation region.

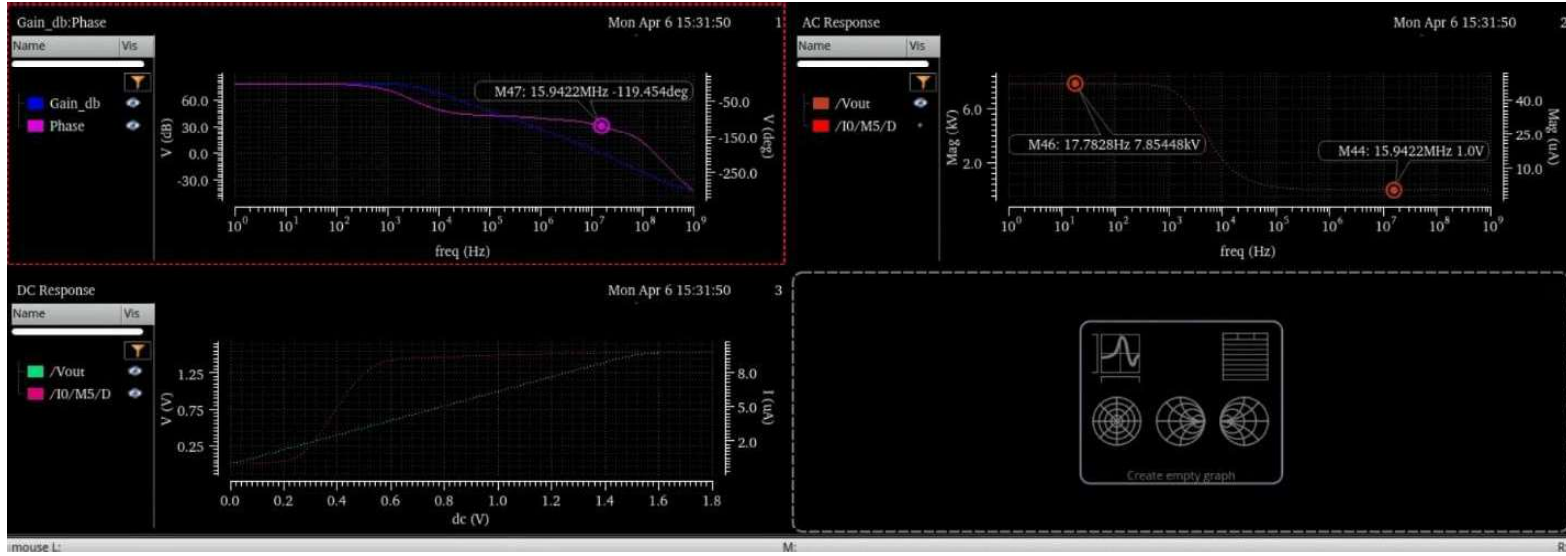


Figure 5: Comprehensive Performance Verification of the Optimized Op-Amp Design: (Top-Left) AC Frequency Response and Phase Margin, (Top-Right) Linear Gain Magnitude, and (Bottom-Left) DC Transfer Characteristics

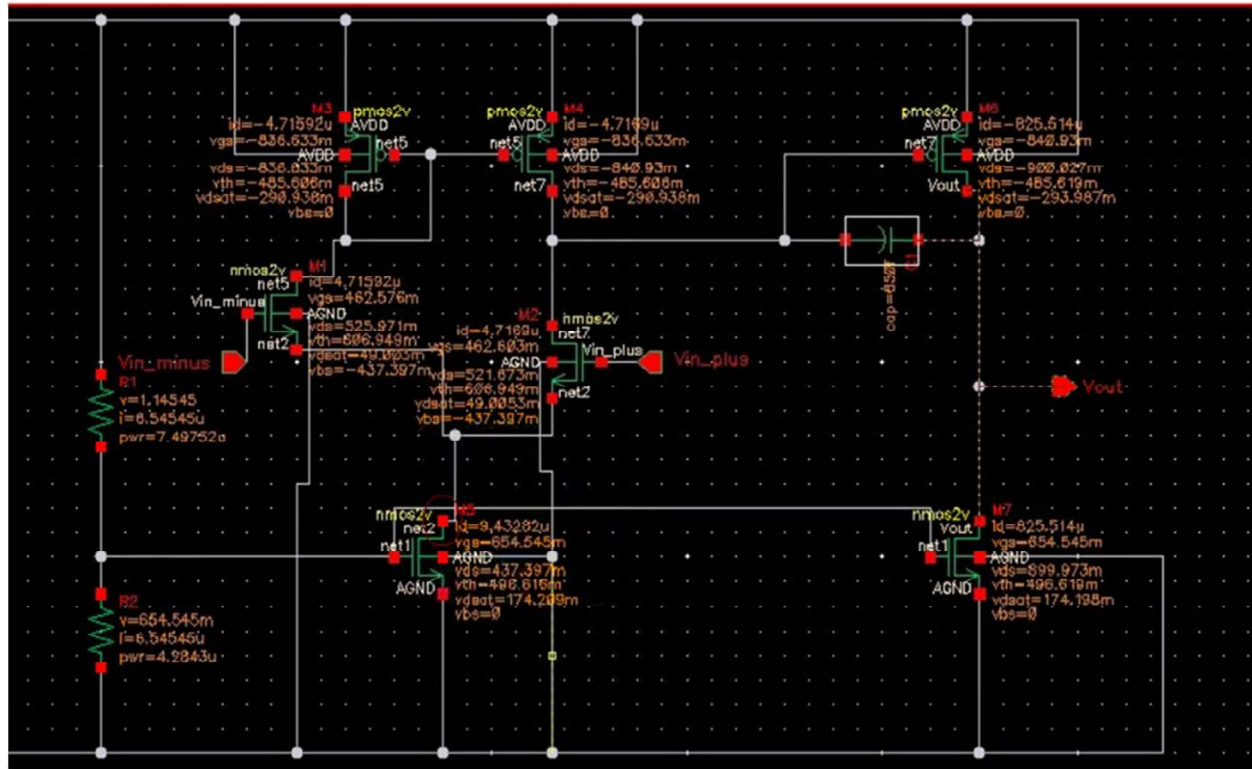


Figure 6: DC Operating Points of MOS Devices

Power Consumption

The total current drawn from the 1.8 V supply was measured under nominal conditions:

- **Total Supply Current (I_P):** 841.49 μA (See Figure 2)
- **Total Power Dissipation:**

$$P = V_{DD} \cdot I_P = 1.8 \text{ V} \cdot 841.49 \mu\text{A} = 1.51 \text{ mW}$$

Large Signal Transient Response

The slew rate was calculated from the transient response using the standard 10%–90% method. The transition time (Δt) was measured as 38 ns for a 0.4 V output change:

$$SR = \frac{\Delta V}{\Delta t} = \frac{1.15 \text{ V} - 0.75 \text{ V}}{38 \text{ ns}} = 10.53 \text{ V}/\mu\text{s}$$

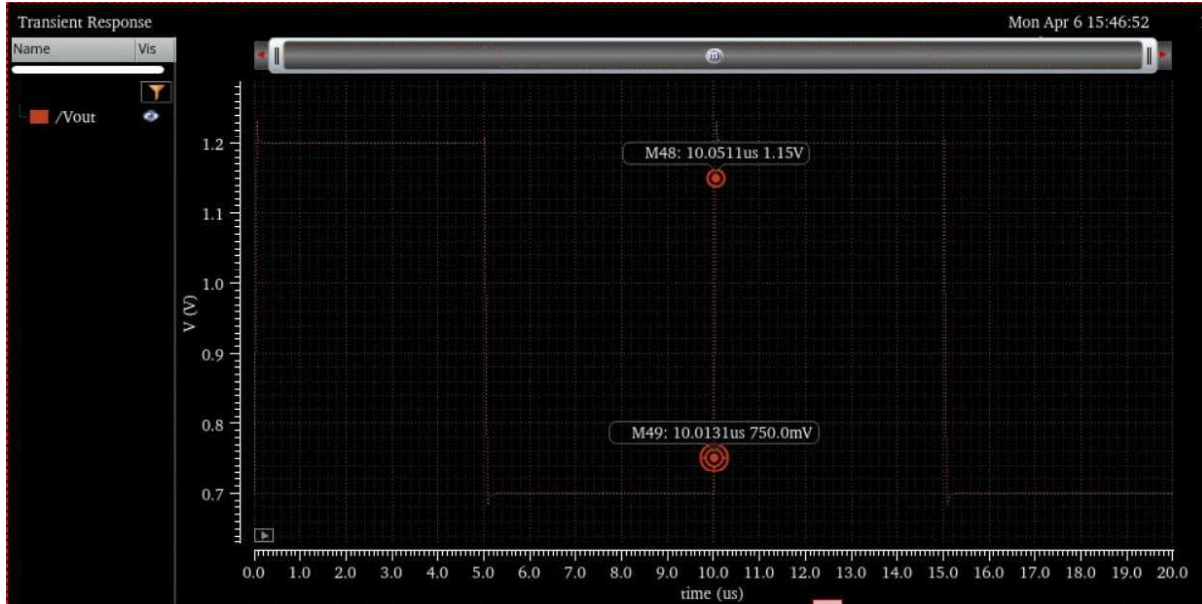


Figure 7: Large-Signal Transient Response showing Slew Rate (10% to 90%)

4. Discussion and Conclusion

(a) Comparison of Design Parameters

The following table compares the transistor dimensions obtained from the second iteration of hand calculations with the final values optimized through simulation.

Transistor	Hand Calc (W/L)	Simulation (W/L)
M _{1,2}	130/1	130/1
M _{3,4}	1/1	1/1
M ₅	2/1	2/1
M ₆	171/1	171/1
M ₇	170/1	170/1

Table 3: Comparison of Transistor Dimensions

(b) Performance Summary

The table below summarizes the design specifications alongside the corresponding simulation results. All performance targets were successfully met.

Parameter	Specification	Simulation Result	Status
Open-Loop Gain	≥ 60 dB	77.9 dB	Pass
Unity-Gain BW	≥ 10 MHz	15.94 MHz	Pass
Phase Margin	$\geq 60^\circ$	60.55°	Pass
Slew Rate	≥ 10.00 V/ μ s	10.53 V/ μ s	Pass
I _p	≤ 5 mA	0.84 mA	Pass

Table 4: Comparison of design specifications Vs. Simulation

(c) Discussion on Optimization and Discrepancies

The transition from hand calculations to high fidelity simulation software required several key optimizations:

- **Bias Network Adjustment:**

The initial hand-calculated value of R_1 was 212 k Ω , corresponding to a bias current of 5 μ A. However, simulation results indicated that parasitic effects and short-channel behavior necessitated a slightly higher current to achieve the target slew rate of 10 V/ μ s. As a result, R_1 was *iteratively* reduced to 175 k Ω . This adjustment improved the slew rate while maintaining power consumption well within the 5 mA limit.

- **Stability Recovery:**

The increase in bias current successfully improved speed but also extended the Unity-Gain Bandwidth (GBW) to 16.75 MHz, which initially reduced the phase margin to slightly below 60°. To restore stability, the Miller compensation capacitor (C_c) was increased from 0.5 pF to 0.65 pF. This modification resulted in a final phase margin of 60.55°, satisfying the stability requirement.

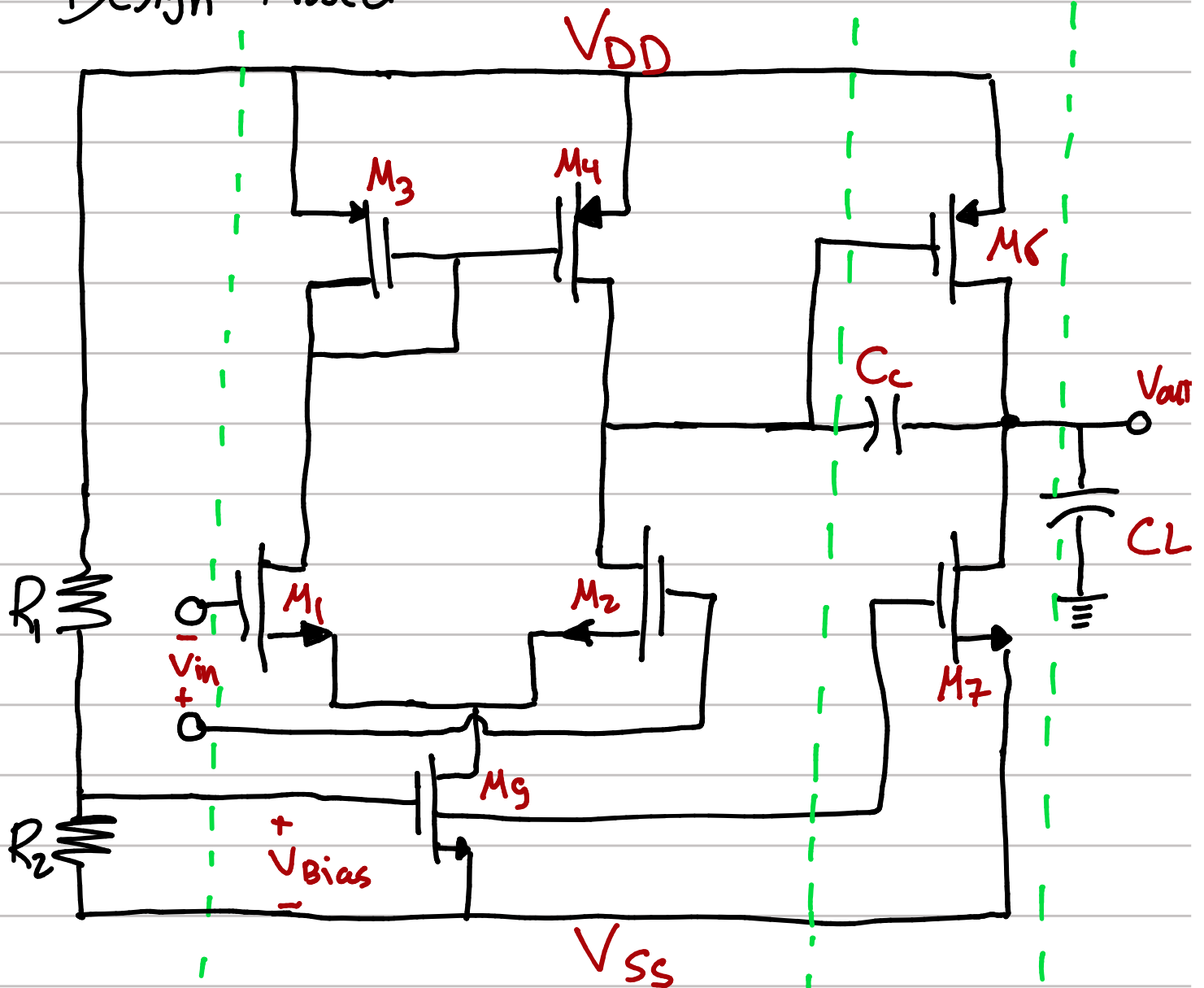
- **Gain Discrepancy:**

The simulated gain (77.9 dB) exceeds the hand-calculated value (60.8 dB) by approximately 17 dB. This difference arises from the simplified square law model used in hand calculations, particularly in estimating the output resistance (r_o). The simulation more accurately captures device behavior, including improved output resistance characteristics in the 0.18 μ m process at the chosen bias conditions.

5. Conclusion

The final operational amplifier design meets or exceeds all specified performance requirements. By employing a two-stage architecture with Miller compensation and an optimized resistive bias network, the circuit achieves an effective balance between speed, gain, and stability. Specifically, the design attains a slew rate of 10.53 V/ μ s, a gain of 77.9 dB, and a phase margin of 60.55°. In addition, the circuit demonstrates strong power efficiency, consuming only 0.84 mA

Design Project:



Biasing
Network

First Stage:
Differential Amplifier

Second Stage: CS Amp
Output

Design Procedure (Hand Calculations): First Iteration

① Given Phase margin $\phi \geq 60^\circ$ and assuming $Z_n \approx 10$ GB: $C_c \geq 0.22 C_L$

$$\therefore C_c \geq (0.22)(2) \text{ pF}$$

$$C_c \geq 0.44 \text{ pF} \quad \therefore \text{Choose } C_c \text{ as } 0.5 \text{ pF}$$

② Given Slew Rate ϕ $10 \text{ V}/\mu\text{s}$, Design $I_{S(\min)}$

$$I_S = \text{SR} \cdot C_c$$

$$= 10 \cdot 0.5$$

$$= 0.5 \times 10^{-5} \text{ A}$$

$$= 5 \times 10^{-6} \text{ A} = 5 \text{ nA}$$

$$I_S = I_{\text{Ref}} \quad \text{Since this is a Current Mirror.}$$

③ Design for S_3 ($S = \frac{W}{L}$) from the max. input Voltage Specification.

$$S = \frac{I_S}{K'_3 [V_{DD} - V_{in(\max)} - |V_{T03}| + V_{T1}]^2}$$

$$K'_3 = \mu_{n,p} \times C_{ox,p} = \mu_{n,p} \times \frac{\epsilon_{ox}}{t_{ox,p}} = 0.0139 \times \frac{3.9 \times 8.85 \times 10^{-14}}{4.05 \times 10^{-9}}$$

$$= 118.17 \text{ nA} / \text{V}^2$$

$$S_3 = \frac{5 \mu A}{118.17 [1.8 - 1.4 - |-0.459| + 0.4395]^2}$$

$$S_3 = 0.29225 = S_4$$

↳ This is the minimum Ratio needed. We Can design $W_3 = W_4 = 1 \mu m$ and $L_3 = L_4 = 1 \mu m$ (Ratio = $1 \ll \underline{1000}_{spec}$)

④ Verification That the Pole of M_3 Will not be dominant

$$GB = 10 \text{ MHz}, \quad \frac{g_{m3}}{2C_{gs3}} > 10 \text{ GB}$$

$$\therefore \frac{g_{m3}}{2C_{gs3}} > 100 \text{ MHz}$$

$$= \frac{\sqrt{2K'_3 S_3 I_3}}{2(0.67 \mu_3 L_3 C_{ox})} \rightarrow I_3 = \frac{I_5}{2}$$

$$= \frac{\sqrt{2(118.17)(1)(2.5)}}{2(0.67(1)(1) \frac{3.9 \times 8.85 \times 10^{-12}}{4.06 \times 10^{-9}})}$$

$$= 2.133791484 \times 10^9 \text{ rad/s}$$

$$\hookrightarrow \times \frac{1}{2\pi} = 340 \text{ MHz} \gg 100 \text{ MHz}$$

Therefore P_3 is not of concern.

⑤ Design for $S_1(S_2)$ to achieve desired GB

$$g_{m1} = GB \cdot C_c, S_1 = \frac{g_{m1}^2}{K_{1,n} I_{D1}}$$

$$g_{m1} = g_{m2} = 10 \text{ MHz} \times 2\pi \times 0.5 \text{ pF} = 3.14 \times 10^{-5} \text{ S}$$

$$K_{1,n} = W_{0,n} \times \frac{E_{ox}}{t_{ox,n}}$$

$$0.0305 \times \frac{3.9 \times 8.85 \times 10^{-12}}{3.981 \times 10^{-9}} = 2.644 \times 10^{-4}$$

$$S_1 = \frac{(3.14 \times 10^{-5})^2}{2.644 \times 10^{-4} \times (5 \times 10^{-6})}$$

$$S_1 = S_2 = 0.7469 = \left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2$$

↪ Min. Ratio

Design $W_1 = W_2 = L_1 = L_2 = 1 \mu\text{m}$

⑥ Design for S_5 from minimum input voltage

$$\begin{aligned} V_{DS5}(\text{sat}) &= V_{in}(\text{min}) - V_{SS} - \sqrt{\frac{I_5}{\beta_1}} - V_{T1} \\ &= 0.8 - 0 - \sqrt{\frac{5 \mu\text{A}}{1.322 \times 10^{-4}}} - 0.4395 \\ &= 0.1658 \text{ V} > 100 \text{ mV} \end{aligned}$$

$$\begin{aligned} \text{Where } \beta_{11} &= \frac{1}{2} \mu_n C_{ox} \cdot S_1 \\ &= \frac{1}{2} (2.644 \times 10^{-4}) \times 1 = 1.322 \times 10^{-4} \text{ uA/V}^2 \end{aligned}$$

$$S_5 = \frac{2I_5}{K'_5 (V_{DS5}(sat))^2}$$

$$= \frac{2(5 \times 10^{-6})}{2.644 \times 10^{-4} (0.1658)^2}$$

$$= 1.3758$$

↳ min Ratio, make $S_5 = 2$
 $W_5 = 2, L_5 = 1$

⑦ Find S_6

Assume $V_{SG4} = V_{SG6}$

$$g_{m6} = 2.2 g_{m2} \left(\frac{C_L}{C_C} \right)$$

$$= 2.2 \times 3.14 \times 10^{-5} \times \frac{2 \text{ pF}}{0.5 \text{ pF}} = 2.7632 \times 10^{-4} \text{ S}$$

↳ This is min, so we would be
 Cutting it close

$$\text{Let } g_{m6} = 10 g_{m2}$$

$$= 3.14 \times 10^{-4} \text{ S}$$

$$S_6 = S_4 \frac{g_{m6}}{g_{m4}} = 1 \times \frac{3.14 \times 10^{-4} \text{ S}}{2.43047 \times 10^{-5} \text{ S}}$$

$$\left(\frac{W}{L} \right)_6 = 12.91 \approx 13$$

$$W_6 = 13 \text{ nm}, L_6 = 1 \text{ nm}$$

8. Calculate I_6

$$I_6 = \frac{g_{m6}^2}{2K_{p6}S_6} = \frac{3.14 \times 10^{-4}}{2 \times (118.17 \times 10^{-6}) \times 13}$$
$$= 3.209 \times 10^{-5} \text{ A}$$

Verify that S_6 satisfies $V_{out}(\max)$

$$V_{out}(\max) = V_{DD} - |V_{ov_6}|$$
$$= 1.8 - \sqrt{\frac{2 \cdot I_6}{K_{p6} S_6}}$$

$$= 1.8 - \sqrt{\frac{2(3.209 \times 10^{-5})}{118.17 \times 10^{-6} \times 13}}$$

$$= 1.595 \text{ V}$$

→ 1.5 V (output swing) so S_6 satisfies $V_{out}(\max)$

9. Since we have all our variables we can calculate Gain

$$A_v = \frac{2g_{m2}g_{m6}}{I_5(\lambda_n + \lambda_p)^2 I_6}$$
$$= \frac{2(3.14 \times 10^{-5})(3.14 \times 10^{-4})}{5 \times 10^{-6}(0.491 + 0.719)^2(3.21 \times 10^{-5})} = 84 \text{ V/V}$$

↳ This Voltage Gain doesn't meet our goal. Thus, we need to increase our g_{m2} and g_{m6} by increasing $S_2 = \left(\frac{W}{L}\right)_2$ and $\left(\frac{W}{L}\right)_6 = S_6$

Second Iteration:

Step 1 - 4 are exactly the same

Step 5:

$$S_1 = S_2 = 0.7459 = \left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2$$

↳ we know that's the min. ratio

let's make $S_1 = S_2 = \left(\frac{W}{L}\right) = 130$
 $W = 130 \text{ nm} = L = 1 \text{ nm}$

$$S_2 = \frac{g_{m2}^2}{K_{2,n} I_S}, \quad \text{Solve for } g_{m2}$$

$$g_{m2} = \sqrt{S_2 K_{2,n} I_S}$$

$$= \sqrt{130 \times 2.644 \times 10^{-4} \times (5 \times 10^{-6})}$$

$$g_{m2} = 4.1456 \times 10^{-4} \text{ S} = g_{m1}$$

Step 6: Design for S_5 from minimum input voltage

$$\begin{aligned}\beta_{n1} &= \frac{1}{2} \mu_n C_{ox} \cdot S_1 \\ &= \frac{1}{2} (2.644 \times 10^{-4}) \times 130 = 0.017186 \text{ A/V}^2\end{aligned}$$

$$\begin{aligned}V_{DS_5}(\text{sat}) &= V_{in}(\text{min}) - V_{SS} - \sqrt{\frac{I_S}{\beta_1}} - V_{T_1} \\ &= 0.8 - 0 - \sqrt{\frac{5 \mu\text{A}}{0.017186}} - 0.4395 \\ &= 0.3434 \text{ V} \gg 100 \text{ mV} \checkmark\checkmark\end{aligned}$$

↳ That means we have a better voltage "budget" for M_5 , the minimum required size for M_5 decreases significantly but we will keep it the same as the first iteration.

$$S_5 = 2$$

Step 7: Find S_6

Assume $V_{SG4} = V_{SG6}$

$$g_{m6} = 2.2 g_{m2} \left(\frac{C_L}{C_C} \right) \\ = 2.2 \times 4.1456 \times 10^{-4} \times \frac{2 \text{ pF}}{0.5 \text{ pF}} = 3.6421 \times 10^{-3} \text{ S}$$

↳ This is minimum, so we would be cutting it close

$$\text{Let } g_{m6} = 10 g_{m2} \\ = 4.1456 \times 10^{-3} \text{ S}$$

$$S_6 = S_4 \frac{g_{m6}}{g_{m4}} = 1 \times \frac{4.1456 \times 10^{-3} \text{ S}}{2.43047 \times 10^{-5} \text{ S}}$$

$$\left(\frac{W}{L} \right)_6 = 170.5 \approx 171$$

$$W_6 = 171 \text{ nm}, L_6 = 1 \text{ nm}$$

8. Calculate I_6

$$I_6 = \frac{g_{m6}^2}{2 K_p' S_6} = \frac{(4.1456 \times 10^{-3})^2}{2 \times (118.17 \times 10^{-6}) \times 171} \\ = 425.23 \text{ nA}$$

Verify that S_S satisfies $V_{out}(\max)$

$$V_{out}(\max) = V_{DD} - |V_{ov_6}|$$
$$= 1.8 - \sqrt{\frac{2 \cdot I_6}{K_P \cdot S_6}}$$

$$= 1.8 - \sqrt{\frac{2 \times 425.23 \times 10^{-6}}{118.17 \times 10^{-6} \times 171}}$$

$$= 1.5948 \text{ V} \geq 1.5 \text{ V} \checkmark \checkmark$$

9. Design S_7 to achieve the desired Current ratio between I_5 and I_6

$$S_7 = \frac{I_6}{I_5} S_5 = \frac{425.23 \mu\text{A}}{5 \times 10^{-6}} \times 2$$
$$= 170$$

$$V_{out, \min} = V_{SS} + |V_{ov_7}|$$
$$0 \text{ V} + \sqrt{\frac{2 \cdot I_7}{K'_n \cdot S_7}}$$

$$= 0 + \sqrt{\frac{2 (425.23 \mu\text{A})}{2.644 \times 10^{-4} \times 170}}$$

$$= 0.137 \text{ V} \leq 0.4 \text{ V} \checkmark \checkmark$$

10. Check Gain

$$\begin{aligned} A_v &= \frac{2 g_{m2} g_{m6}}{I_S (\lambda_n + \lambda_p)^2 I_G} \\ &= \frac{2 (4.1456 \times 10^{-4}) (4.1456 \times 10^{-3})}{(5 \times 10^{-6}) (0.491 + 0.719)^2 (425.23 \times 10^{-6})} \\ &= 1104 \text{ V/V} \quad \checkmark \checkmark \end{aligned}$$

11. Bias Network Design

$$\begin{aligned} V_{ov5} &= \sqrt{\frac{2 I_S}{K'_n \cdot S_S}} \\ &= \sqrt{\frac{5 \times 10^{-6} \times 2}{2.644 \times 10^{-4} \times 2}} = 0.1375 \text{ V} \end{aligned}$$

$$\begin{aligned} V_{ov} &= V_{G_S} - V_{th,n} \\ V_G &= V_{ov} + V_{th,n} \\ &= 0.1375 + 0.4395 \\ &= 0.5770 \text{ V} \end{aligned}$$

$$V_G = V_{DD} \frac{R_2}{R_1 + R_2}$$

Let's say R_2 is $100\text{ K}\Omega$

$$\therefore 0.5770 = 1.8 \frac{100\text{ k}}{R_1 + 100\text{ k}}$$

$$R_1 = 211960\ \Omega = 212\text{ K}\Omega$$

11. Total DC Current drawn from Power Supply

$$I_+ = I_S + I_G + I_{\text{Bias}}$$

$$= (5 \times 10^{-6}) + (425.23 \times 10^{-6}) + \frac{1.8}{212\text{ K}\Omega + 100\text{ K}\Omega}$$

$$= 4.3599 \times 10^{-4}\text{ A}$$

$$\hookrightarrow \leq 5\text{ mA} \quad \checkmark \checkmark$$